

File management:

- Create a file / Modify a file
 - o Touch
 - Create a blank file.
 - Updates the timestamp (last modified date & time) of an existing file.
 - o Cat
 - Create a file
 - View content of the file
 - Override the data/content in a file
 - Append the content to a file
 - Concatenate 2 files data into another file
 - o Nano
 - Useful in Ubuntu OS
 - It's a cmd line editor for create, viewing and modifying a file.
 - o Vi/Vim editor
 - It's a powerful tool to create, viewing and modifying a file.
 - o Gedit editor
 - It's a graphical editor, similar to NOTEPAD in windows.
- Delete a file
 - o `rm <filename>` → delete a specific file
 - o `rm *.txt` → delete files with specific (txt) extension.
- Copy a file
 - o `cp <what-to-copy/source file> <where-to-paste/destination location>`
- Move a file
 - o `mv <source> <destination>` → cut-paste a file
 - o `mv <old-file-name> <new-file-name>` → rename a file
- Hide a file
 - o `mv <old-file-name> .<old-file-name>`
- Unhide a file
 - o `mv .<old-file-name> <old-file-name>`

Directory management

- Create a directory
 - o `mkdir` → Make Directory
- Change the directory
 - o `cd` → Change directory
- Remove a directory
 - o `rmdir` → removes only empty
 - o `rm` → deletes directory as well.

User management:

- Create a user
 - o command → useradd / adduser
- Modify a user
 - o command → usermod
- Delete a user
 - o command → userdel

Whenever, you create a user in linux some files and locations get modified.

- /etc/passwd → file
- /etc/shadow → file
- /etc/groups → file
- /etc/gshadow → file
- /home → location
- /var/spool/mail → location

Task

- Create a user with login name as “rocky”.
- User ID for this user will be 5000
- Comment for this user will be “Italian Stallion”.
- IMP – You cannot use “usermod” command.
- Hint – “man useradd”.

/etc/passwd file:

- ✓ This file contains all the user related information in it.
- ✓ It contains the information in 7 columns, separated by ‘:’.
 1. Username/login name (with which user logs in to the server).
 2. Referenced password (actual password is stored in /etc/shadow).
 3. User ID
 4. Group ID
 5. Comment / description / full name → chfn
 6. Default home directory of the user.
 7. Default shell allocated to that user.
- ✓ Note – do not change anything directly from the file, use “usermod” command to any change.

/etc/shadow file:

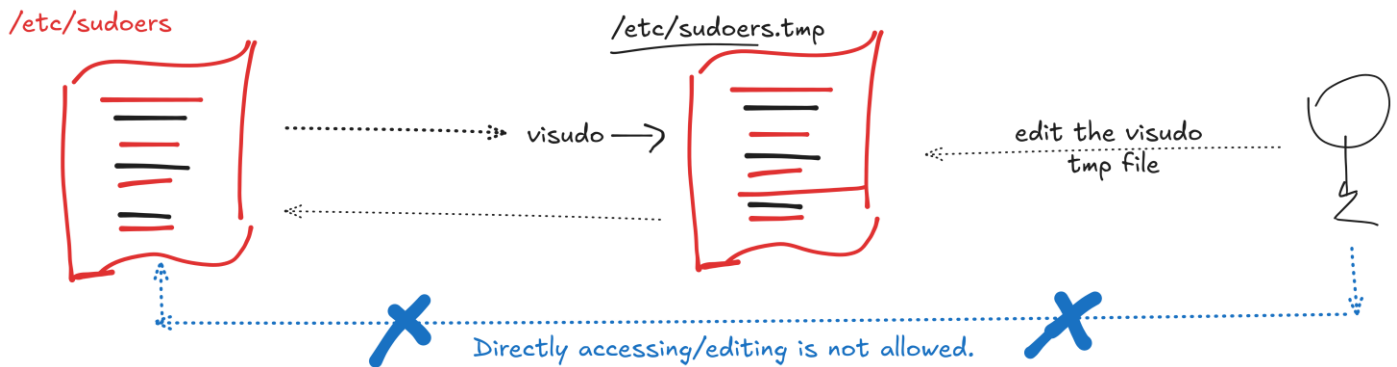
- ✓ This file contains the Encrypted password and password policies in 9 columns.
- ✓ This file contains:
 1. Login name
 2. True Encrypted password
 3. Last password changes since 1st Jan 1970 (epoch date) → 5-digit number
 4. Minimum password age → 0
 5. Maximum password age → 99999 (~273 years) → 60 days
 6. Warning days → 7 days
 7. Inactive days
 8. Account expiry
 9. Unused till date, kept for future purpose.
- ✓ Note – do not change anything directly from the file, use “chage” command to any change.

Sudo user management:

Sudo user = “regular user (\$)” + “limited root privileges”

To make a sudo user:

- File: **/etc/sudoers**
 - o This is a READ-ONLY file.
- Command: **visudo**
 - o This cmd creates a temporary copy of /etc/sudoers file, and we modify this temp file and then the valid changes will be stored in /etc/sudoers file.



To create a sudo user:

1. Allocating a user with FULL ADMIN (root equivalent) access.
 - ✓ Superman **ALL=(ALL)** **ALL**
 - i. **ALL** = terminal on which user must execute the command.
 - ii. **ALL** = which user/group can be access by superman.
 - iii. **ALL** = which command superman can execute.
2. Allocating a user with PRE-DEFINED set of permissions.
 - ✓ Hulk **ALL=(ALL)** **NETWORKING**
3. Allocating a user with LIMITED permissions.
 - ✓ Rocket **ALL=(ALL)** **CUSTOM**